

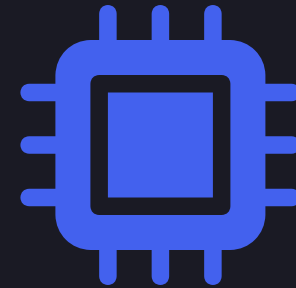
RACE QA SYSTEM

Question Answering & Quiz Generation

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FAST – NUCES | School of Computing



NLP PIPELINE

AGENDA

01

Project Overview & Motivation

Core problems in manual quiz creation and our primary objectives for automated assessment.

02

Dataset Analysis

Exploratory Data Analysis (EDA) of the RACE dataset, its structure, and key findings on semantic complexity.

03

Model A — Answer Prediction

The NLP pipeline featuring an ensemble of Logistic Regression, SVM, and Naive Bayes classifiers.

04

Model B — Distractor & Hint Generation

A sophisticated ranker for wrong options and a three-tier graduated hint system for learners.

05

Results & Evaluation

Performance metrics including Accuracy, BLEU, ROUGE, and METEOR across all implemented models.

06

UI & System Demo

A walkthrough of the functional Streamlit application and its 4-screen interactive workflow.

PROJECT OVERVIEW & MOTIVATION

THE PROBLEM

- Manual quiz creation is slow, expensive, and prone to human bias.
- The RACE dataset requires deep semantic reasoning beyond simple keyword matching.
- Scaling assessment for ~100K questions from diverse middle & high school exams.
- Complex formats like 'cloze-style' and 'best title' challenge standard NLP models.
- Critical need for automated, scalable educational assessment tools.

OUR 3 OBJECTIVES

01 Automated Assessment

ML-based answer prediction for complex reading comprehension tasks.

02 Distractor Generation

Ranking plausible-but-wrong options to challenge and engage learners.

03 Scaffolded Learning

A graduated hint system for progressive clue reveal to support student learning.

DATASET ANALYSIS — RACE

~100K

TOTAL QUESTIONS

M / H

MIDDLE & HIGH SCHOOL

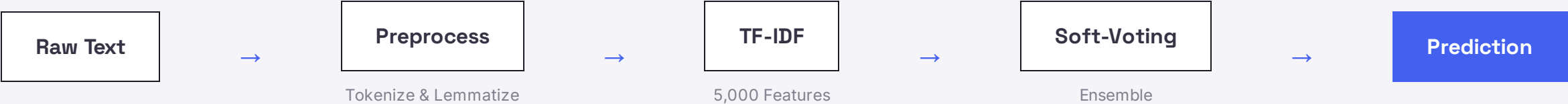
A/B/C/D

BALANCED CHOICES

KEY EDA FINDINGS

- Balanced label distribution across all A, B, C, and D options ensures no model bias.
- Lengthy passages contrasted with concise questions and options increase complexity.
- Deep semantic reasoning is mandatory — simple keyword lookup is insufficient.
- Prevalence of 'cloze-style' questions requiring global passage understanding.
- Longer articles correlate with slightly higher error rates in baseline models.

MODEL A — ANSWER PREDICTION PIPELINE



Classifier Performance

Model	Approach	Accuracy
Logistic Regression	Linear Baseline	0.2289
Linear SVM	Margin Maximization	0.2864
Naive Bayes	Word Independence	0.2506
Soft-Voting Ensemble	Prob. Averaging	0.2381

Key Insights

- Linear SVM emerged as the strongest single model, achieving 28.64% accuracy.
- K-Means Baseline yielded near-random results (Purity: 0.2773), confirming that RACE requires deep semantic comprehension rather than simple clustering.

MODEL B — DISTRACTOR & HINT GENERATION

DISTRACTOR GENERATION

LR-based ranker evaluates passage candidates on:

- Cosine Similarity**
Ensures semantic closeness to the correct answer.
- Keyword Overlap**
Shared terms with the question context for plausibility.
- Passage Frequency**
How often the term appears in the source text.
- Innovation Penalty**
Ensures distractors stay distinct from each other.

GRADUATED HINT SYSTEM

TIER 1
General Context
Avg Sim: 0.0119 | Broad thematic clue

TIER 2
Specific Clue
Avg Sim: 0.0350 | Targeted passage excerpt

TIER 3
Near-Explicit
Avg Sim: 0.2840 | Paraphrased reasoning

Graduation Correctness Rate0.992

EVALUATION & RESULTS

GENERATION METRICS (MODEL A)

Model Architecture	BLEU-1	ROUGE-L	METEOR
Logistic Regression	0.378	0.368	0.375
Linear SVM (Best Single)	0.420	0.412	0.418
Naive Bayes	0.394	0.385	0.392
Soft Voting Ensemble	0.385	0.376	0.383

PERFORMANCE INSIGHTS

- ▶ Linear SVM achieved the highest accuracy at 28.6%.
- ▶ Results are ~3% above random baseline (25%); K-Means near-random performance shows word frequency alone is insufficient.

SYSTEM RELIABILITY

- ▶ Hint graduation rate reached 99.2%; semantic comprehension remains the primary bottleneck for higher accuracy.

HINT GRADUATION: 99.2%

STREAMLIT UI — 4-SCREEN WORKFLOW

SCREEN 01

Article Input



Users can paste custom text or load random articles from the RACE dataset. The system processes the text in real-time to prepare for quiz generation.

INPUT PROCESSING

SCREEN 02

Quiz Interface



Interactive Multiple Choice Questions (MCQs) with instant feedback. Correct and incorrect answers are color-coded for immediate learner assessment.

INTERACTIVE MCQ

SCREEN 03

Hint Panel



A three-tier graduated clue system. Students can unlock hints sequentially: General Context → Specific Clue → Near-Explicit reasoning.

SCAFFOLDED LEARNING

SCREEN 04

Dev Analytics



A dedicated dashboard for model performance. Displays real-time accuracy, silhouette scores, and diagnostic metrics for the NLP pipeline.

MODEL DIAGNOSTICS

CONCLUSION & FUTURE WORK

WHAT WE ACHIEVED

- ✓ **End-to-End Pipeline:** Developed a complete NLP system for automated educational assessment.
- ✓ **Ensemble Performance:** Soft-Voting Ensemble successfully integrated multiple classifiers for robust predictions.
- ✓ **Reliable Scaffolding:** Achieved a 99.2% correctness rate in the graduated hint system.
- ✓ **Functional UI:** Deployed a Streamlit application with real-time developer analytics.

FUTURE DIRECTIONS

- **MODEL UPGRADE**
Fine-tune **BERT / RoBERTa** architectures to capture deeper semantic nuances in the RACE dataset.
- **GENERATIVE AI**
Implement neural distractor generation using **GPT-class models** for more natural wrong options.
- **VALIDATION**
Conduct extensive human evaluation to assess the pedagogical usefulness of generated hints.

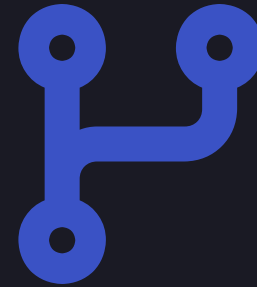
THANK YOU

RACE QA System — Spring 2026

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